Statistical Methodology for Celestial Object Classification

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1 Statistical Methodology

1.1 Overview

The classification of celestial objects such as stars, galaxies, and quasars presents a practical statistical challenge due to overlapping observational properties and the inherent variability of astronomical data. Differences in measurement scales, observational noise, and incomplete information make it difficult to separate object types using simple analytical rules. Therefore, a structured statistical approach is needed to convert raw data into meaningful, comparable, and interpretable forms suitable for modeling.

The analytical process developed in this study follows a sequential logic. First, data preparation ensures quality and consistency, reducing errors that could influence later stages of analysis. Second, feature standardization converts variables measured on different scales into a uniform range, allowing fair and balanced model training. Next, exploratory data analysis (EDA) is carried out to understand how individual features behave, how they relate to each other, and how the three object classes differ in feature space. Based on these insights, classification models are developed to identify patterns that can distinguish between object types. Finally, the results are evaluated to confirm that the models perform consistently and that the classifications align with known astronomical characteristics.

1.2 Data Structure and Problem Context

The dataset, sourced from large-scale astronomical surveys such as the Sloan Digital Sky Survey (SDSS), comprises n observations, each characterized by p features derived from photometric and spectroscopic measurements. These features include apparent magnitudes across optical and near-infrared filters (u, g, r, i, z), redshift as a cosmological distance indicator, and astrometric coordinates (right ascension and declination). The data is represented as a matrix:

$$X = \{x_{ij}\}_{n\times p}, \quad i = 1, 2, \ldots, n;\; j = 1, 2, \ldots, p,$$

where x_{ij} denotes the j-th feature of the i-th observation. The corresponding response vector contains class labels:

$$y = (y_1, y_2, \ldots, y_n)^\top, \quad y_i \in \{\text{Star}, \text{Galaxy}, \text{Quasar}\}.$$

The objective is to estimate a classification function:

$$f : \mathbb{R}^p \to \{1, 2, 3\},$$

that assigns each feature vector x_i to its most probable class while minimizing misclassification risk. This assumes observations are independent and identically distributed (i.i.d.) from a joint distribution $P(X, Y)$, and the learned function f must generalize effectively to unseen data, accommodating challenges like class imbalance and feature heterogeneity.

1.3 Data Preprocessing and Standardization

Astronomical data are inherently complex, with measurements encompassing diverse scales (e.g., magnitudes vs. angular coordinates), potential missing entries due to observational constraints, and outliers from cosmic phenomena or instrumental errors. Without careful preprocessing, these issues can bias statistical inference and degrade model performance. The preprocessing phase is thus designed to ensure data integrity and model readiness through two critical steps.

First, **cleaning and screening** establishes a reliable dataset. Non-informative identifiers (e.g., catalog IDs like ‘objid‘ or ‘specobjid‘) are removed, as they lack predictive value and may introduce spurious correlations. Missing values are assessed column-wise using summary statistics (e.g., null counts) and handled based on their nature: sparse, random missingness imputed using median values to preserve data volume, while systematic gaps prompt record exclusion to avoid bias. Outliers are identified using robust statistical measures, such as the interquartile range (IQR) rule, where values outside $[Q_1 - 1.5 \times \text{IQR}, Q_3 + 1.5 \times \text{IQR}]$ are flagged. These are cross-validated with visual inspections (e.g., boxplots) to distinguish measurement errors (e.g., negative magnitudes) from valid astronomical anomalies (e.g., high-redshift quasars).

Second, **feature standardization** normalizes features to ensure equitable contributions to modeling. Each feature is transformed to zero mean and unit variance:

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j},$$

where $\bar{x}_j = \frac{1}{n}\sum_{i=1}^{n} x_{ij}$ is the sample mean and $s_j = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_{ij} - \bar{x}_j)^2}$ is the sample standard deviation. This step is essential for distance-based methods like k-Nearest Neighbors, where unnormalized features could skew Euclidean distances, and for tree-based methods, where scale disparities may affect split thresholds. Standardization ensures that all features, whether magnitudes, redshifts, or coordinates, contribute proportionally, enhancing model stability and interpretability.

1.4 Exploratory Data Analysis

Exploratory Data Analysis (EDA) is a pivotal step that elucidates the statistical and physical structure of the dataset, informing model design and validating astronomical expectations. By systematically examining feature distributions, relationships, and class characteristics, EDA bridges raw data and predictive modeling.

The process begins with **univariate analysis**, using histograms and boxplots to characterize each feature’s distribution. Histograms reveal shapes—e.g., redshift’s right-skewness due to quasars’ high cosmological values—while boxplots highlight central tendencies, variability, and outliers (e.g., negative magnitudes indicating measurement errors). This step identifies features requiring transformation (e.g., logarithmic scaling for skewed data) and confirms expected patterns, such as stars clustering near zero redshift.

Next, **bivariate and multivariate analyses** explore feature interactions and class separability. Pairwise scatter plots, such as redshift versus u-g color index, visualize clustering tendencies, often revealing distinct regions for stars, galaxies, and quasars. Correlation matrices quantify multicollinearity using the Pearson coefficient:

$$r_{jk} = \frac{\sum_{i=1}^{n}(x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k)}{\sqrt{\sum_{i=1}^{n}(x_{ij} - \bar{x}_j)^2 \sum_{i=1}^{n}(x_{ik} - \bar{x}_k)^2}} = \frac{\text{Cov}(X_j, X_k)}{\sqrt{\text{Var}(X_j)\,\text{Var}(X_k)}}.$$

High correlations among photometric filters (e.g., u and g) suggest redundancy, prompting consideration of dimensionality reduction techniques like PCA. Class-conditional distributions are also examined to validate astronomical insights, such as quasars’ elevated redshift values due to their prevalence in the early universe. This phase not only informs feature selection but also ensures the data aligns with astrophysical principles, grounding the analysis in domain knowledge.

1.5 Model Development

To address the nonlinear boundaries separating celestial classes, three complementary statistical learning models are developed, each capturing distinct aspects of the data. Hyperparameter tuning via grid search with cross-validation ensures optimal performance, balancing bias and variance.

The **k-Nearest Neighbors (kNN)** classifier is a non-parametric method that assigns a test observation x_0 to the majority class among its k nearest neighbors in standardized feature space. Neighbors are identified using Euclidean distance, and the prediction is:

$$\hat{y}(x_0) = \arg\max_{c \in \mathcal{C}} \sum_{i \in \mathcal{N}_k(x_0)} \mathbf{1}(y_i = c),$$

where $\mathcal{N}_k(x_0)$ is the set of indices of the k closest points, and $\mathbf{1}(\cdot)$ is the indicator function. Hyperparameters (e.g., k from 1 to 15, uniform or distance-based weighting) are tuned to optimize local pattern capture, leveraging the data's geometric structure.

The **Decision Tree** classifier constructs a hierarchical model by recursively splitting the feature space based on feature-threshold pairs that maximize purity. The Gini impurity at node t is:

$$G(t) = 1 - \sum_{c=1}^{C} p(c|t)^2,$$

where $p(c|t)$ is the proportion of class c samples. Splits maximize the impurity reduction:

$$\Delta G = G(t) - \sum_{j} \frac{n_j}{n_t} G(t_j),$$

with n_j/n_t as the proportion of samples in child node j. Parameters like maximum depth and minimum samples per leaf are tuned to balance complexity and interpretability, producing rules like "redshift > 1.5 implies quasar."

The **Random Forest** classifier builds an ensemble of T decision trees, each trained on a bootstrap sample with random feature subsampling at splits to decorrelate predictions. The final prediction is:

$$\hat{y}(x) = \text{mode}\{h_t(x)\}_{t=1}^{T},$$

where $h_t(x)$ is the t-th tree's output. Tuning parameters (e.g., number of trees, maximum features) enhances robustness, and feature importance scores highlight key predictors like redshift.

1.6 Model Evaluation and Interpretation

Model evaluation goes beyond accuracy to assess robustness, generalizability, and scientific relevance. A stratified train-test split (e.g., 70%-30%) preserves class proportions, while repeated hold-out validation estimates performance variance. Additionally, k-fold cross-validation (e.g., $k = 5$ or 10) computes average metrics across folds:

$$\widehat{M} = \frac{1}{k} \sum_{j=1}^{k} M_j.$$

Key metrics include:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN},$$
$$\text{Precision} = \frac{TP}{TP + FP},$$
$$\text{Recall} = \frac{TP}{TP + FN},$$

$$\text{F1-score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}.$$

with macro-averaging for multi-class balance. Confusion matrices reveal misclassification patterns, such as quasar-galaxy overlaps due to redshift similarities.

Interpretation is central: kNN captures local clustering, Decision Trees provide explicit rules, and Random Forests quantify feature importance (e.g., redshift's dominance). These insights align with astronomical knowledge, validating the models' physical relevance while balancing predictive power and transparency.